

- * Weigh all patients and track BMI during treatment
- Get baseline personal and family history of diabetes, obesity, dyslipidemia, hypertension, and cardiovascular disease
- * Get waist circumference (at umbilicus), blood pressure, fasting plasma glucose, and fasting lipid profile
- Determine if the patient
 - is overweight (BMI 25.0–29.9)
 - is obese (BMI >30)
 - has pre-diabetes (fasting plasma glucose 100–125 mg/dL)
 - has diabetes (fasting plasma glucose >126 mg/dL)
 - has hypertension (BP >140/90 mmHg)
 - has dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol)
- Treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

Monitoring after starting any atypical antipsychotic

- * BMI monthly for 3 months, then quarterly
- * Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics

* Blood pressure, fasting plasma glucose, fasting lipids within 3 months and then annually, but earlier and more frequently for patients with diabetes or who have gained >5% of initial weight

- Treat or refer for treatment and consider switching to another atypical antipsychotic for patients who become overweight, obese, pre-diabetic, diabetic, hypertensive, or dyslipidemic while receiving an atypical antipsychotic

* Even in patients without known diabetes, be vigilant for the rare but life-threatening onset of diabetic ketoacidosis, which always requires immediate treatment, by monitoring for the rapid onset of polyuria, polydipsia, weight loss, nausea, vomiting, dehydration, rapid respiration, weakness and clouding of sensorium, even coma

- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months, and blonanserin should be discontinued at the first sign of decline in WBC in the absence of other causative factors
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia
- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin
- Mechanism of weight gain and increased incidence of diabetes and dyslipidemia with atypical antipsychotics is unknown, and risks vary based on the particular agent

Notable Side Effects

- Akathisia, drug-induced parkinsonism
- Insomnia, anxiety, sedation
- Urinary retention
- Tardive dyskinesia (TD) (reduced risk compared to conventional antipsychotics)
- Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration



Life-Threatening or Dangerous Side Effects

- Hyperglycemia, in some cases extreme and associated with ketoacidosis or hyperosmolar coma or death, has been reported in patients taking atypical antipsychotics

- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis
- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
- Rare seizures

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait
- Wait
- Wait
- Anticholinergics may reduce drug-induced parkinsonism when present

- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain
- Switch to another atypical antipsychotic

Best Augmenting Agents for Side Effects

- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 8–16 mg/day divided in 2 doses

Dosage Forms

- Tablet 2 mg, 4 mg, 8 mg

- Powder 20 mg per 1 g powder

How to Dose

- Initial 8 mg/day divided in 2 doses; maintenance dose 8–16 mg/day divided in 2 doses; maximum dose 24 mg/day
- Blonanserin should be taken after a meal, as maximum concentrations are increased in the fed state; however, because the increase in systemic exposure continues until at least 4 hours after food intake, blonanserin can be taken before bedtime



Dosing Tips

- Start with twice daily dosing; once stabilized, some patients do well with 1 dose given at night
- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly
- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine
- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- Treatment should be suspended if absolute neutrophil count falls below $1000/\text{mm}^3$

Overdose

- Limited data

Long-Term Use

- Not extensively studied past 56 weeks, but long-term maintenance treatment is often necessary for schizophrenia
- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- Down-titration, especially when simultaneously beginning a new antipsychotic while switching (i.e., cross-titration)
- Rapid discontinuation could theoretically lead to rebound psychosis and worsening of symptoms

Pharmacokinetics

- Elimination half-life 10–16 hours after single dose
- Metabolized by CYP450 3A4



Drug Interactions

- CYP450 3A4 inducers, such as carbamazepine, can lower the plasma levels of blonanserin
- CYP450 3A4 inhibitors, such as ketoconazole, nefazodone, fluvoxamine, and fluoxetine, can increase plasma levels of blonanserin
- May increase effects of antihypertensive agents
- May antagonize levodopa, dopamine agonists

Other Warnings/Precautions

- Use with caution in patients with conditions that predispose to hypotension (dehydration, overheating)
- Atypical antipsychotics have the potential for cognitive and motor impairment, especially related to sedation
- Atypical antipsychotics can cause body temperature dysregulation; use caution in patients who may experience conditions that increase body temperature (e.g., strenuous exercise, saunas, extreme heat, dehydration, or concomitant anticholinergic medication)
- Dysphagia has been associated with antipsychotic use, and blonanserin should be used cautiously in patients at risk for aspiration pneumonia
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic

treatment, and recurrently for patients on long-term antipsychotic therapy

- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- In conjunction with adrenaline/epinephrine
- If patient is taking ketoconazole
- If there is a proven allergy to blonanserin

Special Populations

Renal Impairment

- Not studied

Hepatic Impairment

- Use with caution; may need to lower dose

Cardiac Impairment

- Use in patients with cardiac impairment has not been studied, so use with caution

Elderly

- Some patients may tolerate lower doses better

- Although atypical antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with atypical antipsychotics are at an increased risk of death compared to placebo and also have an increased risk of cerebrovascular events

Children and Adolescents

- Safety and efficacy have not been established
- Children and adolescents using blonanserin may need to be monitored more often than adults and may tolerate lower doses better

Pregnancy

- Controlled studies have not been conducted in pregnant women
- Psychotic symptoms may worsen during pregnancy, and some form of treatment may be necessary

Breast Feeding

- Unknown if blonanserin is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed unless the potential benefit to the mother justifies the potential risk to the child

- Infants of women who choose to breast feed while on blonanserin should be monitored for possible adverse effects

The Art of Psychopharmacology

Potential Advantages

- May be useful when other antipsychotics have failed to provide adequate response or have not been tolerated

Potential Disadvantages

- Patients who require once daily dosing from the initiation of treatment
- Patients who require intramuscular administration

Primary Target Symptoms

- Positive symptoms of psychosis
- Negative symptoms of psychosis
- Cognitive symptoms
- Unstable mood (both depression and mania)
- Aggressive symptoms

Pearls

- Relatively selective binding profile

- Not approved for mania, but almost all atypical antipsychotics approved for acute treatment of schizophrenia have proven effective in the acute treatment of mania as well
- Blonanserin's binding affinity for D2 dopamine receptors (0.14 nM) is greater than for D3 receptors (0.49 nM); however, it is one of the very few antipsychotics in which the binding affinity for D3 receptors is greater than dopamine's binding affinity for D3 receptors (~60 nM)
- D3 antagonist actions have been hypothetically associated with improvement in negative, cognitive, and affective symptoms

Suggested Reading

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Yang J , Bahk WM , Cho HS , et al. Efficacy and tolerability of blonanserin
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compared trial . *Clin Neuropharmacol* 2010 ;**33** (4):169–75 .

Bremelanotide

Therapeutics

Brands

- Vyleesi
- *see index for additional brand names*

Generic?

No

Class

- Nonselective melanocortin receptor agonist

Commonly Prescribed for

(bold for FDA approved)

**Acquired, generalized hypoactive sexual desire disorder (HSDD)
in premenopausal women**

How the Drug Works

- Bremelanotide is a nonselective melanocortin receptor agonist; at therapeutic doses, bremelanotide binding to melanocortin 1 and 4 receptors is most relevant

- Sexual dysfunction is theoretically linked to an imbalance in central excitatory and inhibitory sexual signals
- HSDD hypothetically results from excessive inhibitory signals, inadequate excitatory signals, or a combination of the two
- Melanocortin receptors are located in the medial preoptic area of the hypothalamus, which is implicated in the sexual behavior of both sexes; stimulation of those receptors leads to dopamine release

How Long Until It Works

- Should be used at least 45 minutes before anticipated sexual activity; duration of efficacy after each dose is unknown and optimal window for bremelanotide administration has not been established

If It Works

- Increases ratings on sexual desire scores
- Reduces distress related to sexual dysfunction

If It Doesn't Work

If there is no improvement after 8 weeks, discontinue use



Best Augmenting Combos for Partial Response or Treatment Resistance

- None known

Tests

Blood pressure and risk for cardiovascular disease should be evaluated before initiation and periodically during treatment

Side Effects

How Drug Causes Side Effects

- Binding at melanocortin 1 receptors on melanocytes leads to melanin expression and increased pigmentation

Notable Side Effects

- Nausea
- Flushing
- Injection site reactions
- Headache
- Vomiting
- Hyperpigmentation (higher risk with daily dosing)



Life-Threatening or Dangerous Side Effects

- Transient increase in blood pressure and decrease in heart rate

Weight Gain



- None

Sedation



- Reported but not expected

What to Do About Side Effects

- Wait
- In most cases nausea improves after the first dose; for persistent or severe nausea, consider discontinuation or initiating antiemetic therapy
- For hyperpigmentation, consider discontinuation
- Switch to another treatment option

Best Augmenting Agents for Side Effects

- Antiemetic medication

Dosing and Use

Usual Dosage Range

- 1.75 mg as needed, at least 45 minutes before anticipated sexual activity

Dosage Forms

- Subcutaneous injection 1.75 mg/0.3 mL in a single-dose autoinjector

How to Dose

- Patient should self-inject 1.75 mg subcutaneously in the abdomen or thigh using the autoinjector, as needed, at least 45 minutes before anticipated sexual activity



Dosing Tips

- Do not administer more than one dose in 24 hours
- Not recommended to administer more than 8 doses per month
- Duration of efficacy after each dose is unknown and the optimal window for bremelanotide administration has not been established
- Patient should be advised to visually inspect the drug product prior to administration, and to discard if the solution is cloudy or discolored or if visible particles are observed

Overdose

- Limited data; nausea, focal hyperpigmentation, and more pronounced blood pressure increases are more likely with higher doses

Long-Term Use

- Safe and effective in controlled trials (24 weeks) and open-label extension studies lasting an additional 52 weeks

Habit Forming

- No

How to Stop

- Bremelanotide is administered as needed

Pharmacokinetics

- T_{max} is approximately 1 hour
- Mean terminal half-life approximately 2.7 hours
- Absorption is not affected by injection site (thigh or abdomen)



Drug Interactions

- Bremelanotide may slow gastric emptying, which could impact the rate and extent of absorption of concomitantly administered oral medications
- Bremelanotide may significantly decrease exposure to oral naltrexone, so avoid use with orally administered naltrexone-containing products



Other Warnings/Precautions

- Bremelanotide can cause transient increase in blood pressure and decrease in heart rate, which occurs after each dose and usually resolves within 12 hours; patients should be evaluated for cardiovascular risk before and periodically during treatment; treatment should not be initiated in those at high risk for cardiovascular disease
- Focal hyperpigmentation can occur, with greater risk in patients with darker skin and with daily dosing; resolution was not confirmed in some patients; consider discontinuation if hyperpigmentation develops
- Nausea can be severe enough to require antiemetic therapy or treatment discontinuation; for most patients in clinical trials it improved with the second dose; for persistent nausea, consider discontinuation or initiation of antiemetic therapy

Do Not Use

- In patients with uncontrolled hypertension or known cardiovascular disease
- If there is a proven allergy to bremelanotide

Special Populations

Renal Impairment

- Dose adjustment not necessary for mild to moderate impairment

- Use with caution in patients with severe impairment

Hepatic Impairment

- Dose adjustment not necessary for mild to moderate impairment
- Use with caution in patients with severe impairment

Cardiac Impairment

- Contraindicated

Elderly

- Not approved for use in postmenopausal women, and safety and efficacy have not been established



Children and Adolescents

- Safety and efficacy have not been established



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women

- In clinical trials of up to 12 months, 7 pregnancies were reported; among these 7 pregnancies, no major congenital anomalies were reported; there was one spontaneous abortion (miscarriage), five full-term live births, and one outcome unknown as it was lost to follow-up
- When administered to pregnant dogs during organogenesis, embryofetal toxicity occurred at doses approximately 16 times the maximum recommended human dose (MRHD)
- When administered to pregnant mice during pregnancy and lactation, developmental effects occurred at doses approximately 125 times the MRHD
- Advise patients to discontinue bremelanotide if pregnancy is suspected
- Pregnancy registry for bremelanotide: 1-877-411-2510

Breast Feeding

- Unknown if bremelanotide is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- Bremelanotide is dosed episodically (average 2–3 times per month) and has a half-life of 2.7 hours

The Art of Psychopharmacology

Potential Advantages

- Can be taken as needed
- For patients who do not respond to flibanserin

Potential Disadvantages

- Patients with cardiovascular disease
- Patients taking oral medications for which gastric absorption is relevant (e.g., antibiotics)
- Patients who do not like injections

Primary Target Symptoms

- Reduced sexual desire

Pearls

- Bremelanotide is not approved for treatment in postmenopausal women or in men and is not indicated to enhance sexual performance
- If not effective for HSDD, consider avoiding any concomitant use of agents that either enhance serotonin activity (such as SSRIs or SNRIs) or diminish dopamine activity (such as antipsychotics)
- In women of childbearing potential who are or are likely to become sexually active, should inform about risk of harm to the fetus and monitor contraceptive status
- Patients should be counseled that bremelanotide offers no protection against sexually transmitted diseases

- No studies, but theoretically possible to combine with flibanserin for partial responders to either drug

Suggested Reading

Kingsberg SA , Clayton AH , Portman D , et al. Bremelanotide for the treatment of hypoactive sexual desire disorder: two randomized phase 3 trials . *Obstet Gynecol* 2019 ;**134** (5):899 –908 .

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Brexanolone

Therapeutics

Brands

- Zulresso
- *see index for additional brand names*

Generic?

No

◆ Class

- Neuroactive steroid

Commonly Prescribed for

(bold for FDA approved)

- **Postpartum depression**

⚙️ How the Drug Works

- Positive allosteric modulator at neuroactive steroid sites on both gamma-aminobutyric acid (GABA) A benzodiazepine-sensitive and benzodiazepine-insensitive ligand-gated ion channels
- Benzodiazepine-insensitive GABA-A receptor subtypes (with $\alpha 1$ subunits and $\alpha 2$ or $\alpha 3$ subunits) are located extrasynaptically,